Unit: 6 Unit 8: Quadratics **Name:** ____________
Topic: Solving by Quadratic Formula (Discriminant) **Date:** __________

Foundation (Jalapeno)

Q1. What is the quadratic formula?

(A) $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$

(B) $x = \frac{-b\mp\sqrt{b^2-4ac}}{2a}$

(C) $x = \frac{-b\pm\sqrt{b^2+4ac}}{2a}$

(D) $x = \frac{-b\mp\sqrt{b^2+4ac}}{2a}$

(E) $x = \frac{-b\pm\sqrt{4ac-b^2}}{2a}$

Q2. What does the discriminant $b^2 - 4ac$ represent?

(A) Number of solutions
(B) Type of solutions
(C) Both A and B
(D) Neither A nor B
(E) Solutions to the equation

Q3. If the discriminant is positive, what type of solutions does the equation have?

(A) One real solution
(B) Two real solutions
(C) One complex solution
(D) Two complex solutions
(E) No solutions

Q4. If the discriminant is zero, what type of solutions does the equation have?

(A) One real solution
(B) Two real solutions
(C) One complex solution
(D) Two complex solutions
(E) No solutions

Q5. If the discriminant is negative, what type of solutions does the equation have?

(A) One real solution
(B) Two real solutions
(C) One complex solution
(D) Two complex solutions
(E) No solutions

Q6. Solve for x using the quadratic formula: $x^2 + 5x + 6 = 0$

(A) $x = -2, x = -3$
(B) $x = -2, x = 3$
(C) $x = 2, x = -3$
(D) $x = 2, x = 3$
(E) $x = -1, x = -6$

Q7. Solve for x using the quadratic formula: $x^2 - 4x - 3 = 0$

(A) $x = -1, x = 3$
(B) $x = -1, x = -3$
(C) $x = 1, x = 3$
(D) $x = 1, x = -3$
(E) $x = 3, x = -1$

Q8. Solve for x using the quadratic formula: $x^2 + 2x + 1 = 0$

(A) $x = -1, x = 1$
(B) $x = -1, x = -1$
(C) $x = 1, x = 1$
(D) $x = 1, x = -1$
(E) $x = -1, x = 0$

Open Ended Questions (FRQ)

Q9. Solve for x using the quadratic formula: $2x^2 + 5x + 2 = 0$. Show all work and explain your answer.

Q10. Find the discriminant of the quadratic equation $x^2 - 3x - 2 = 0$. Then, determine the number and type of solutions. Explain your reasoning.

Chef's Tips

- Tip 1: Ensure students understand the quadratic formula and its components.
- Tip 2: Emphasize the importance of the discriminant in determining the number and type of solutions.
- Tip 3: Have students show all work and explain their reasoning for free response questions.